

Title

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Notation

The symbols below distinguish upper-level strategic decisions from lower-level ISO-cleared decisions. Scenario probabilities are denoted by ω_s throughout the model to avoid using π , which is commonly reserved for prices or profits in related market models.

Symbol	Description	Unit
Sets and Indices		
$s \in \mathcal{S}$	Activation scenario index and set.	–
$t \in \mathcal{T}$	Time-period index and set.	–
$g \in \mathcal{G}$	Generator index and set.	–
$n \in \mathcal{N}$	Network node index and set.	–
G_n	Set of generators located at node n .	–
$i \in \mathcal{I}$	BESS investment or storage-unit index and set.	–
$b \in \mathcal{B}$	aFRR market block index and set.	–
$l \in \mathcal{L}$	Transmission-line index and set.	–
$t \in b$	Time periods contained in aFRR block b .	–
Upper-Level Decision Variables		
$X_{i,n}^{power}$	Installed BESS power capacity chosen by the upper-level strategic investor.	MW
$X_{i,n}^{energy}$	Installed BESS energy capacity chosen by the upper-level strategic investor.	MWh
$R_{i,n,b}^{offer}$	Strategic aFRR reserve capacity offered by BESS i at node n in block b .	MW
Lower-Level Decision Variables		
$P_{g,t,s}$	Dispatch of generator g in time period t and scenario s .	MW
$P_{n,t,s}^{shed}$	Load shedding at node n , time t , and scenario s .	MW
$P_b^{aFRR,def}$	Unserved aFRR reserve demand in block b .	MW
$NetInjection_{n,t,s}$	Net nodal injection used in the DC network representation.	MW
$R_{i,n,b}^{aFRR}$	ISO-cleared aFRR reserve capacity from BESS i at node n in block b .	MW
$P_{i,n,t,s}^{charge}$	BESS charging power.	MW
$P_{i,n,t,s}^{discharge}$	BESS discharging power.	MW
$P_{i,n,t,s}^{aFRR,up}$	Upward aFRR activation energy represented as power withdrawal or injection adjustment. ^a	MW

Symbol	Description	Unit
$P_{i,n,t,s}^{aFRR,down}$	Downward aFRR activation energy represented as power withdrawal or injection adjustment. ^a	MW
$SOC_{i,n,t,s}$	State of charge of BESS i at node n , time t , and scenario s .	MWh
Dual Variables		
$\lambda_{n,t,s}$	Dual variable of nodal power balance; nodal energy price.	€/MWh
$\lambda_{sys,t,s}$	Dual variable of system-wide power balance.	€/MWh
$\lambda_{aFRR,b}$	Dual variable of the aFRR reserve-capacity clearing constraint; aFRR capacity price. ^b	€/MW
$\nu_{g,t,s}^-, \nu_{g,t,s}^+$	Dual variables of the lower and upper generator dispatch bounds.	€/MWh
$\mu_{l,t,s}^-, \mu_{l,t,s}^+$	Dual variables of the lower and upper transmission-flow bounds.	€/MWh
$\rho_{i,n,t,s}^-, \rho_{i,n,t,s}^+$	Dual variables of the BESS charging-side power-headroom bounds.	€/MWh
$\sigma_{i,n,t,s}^-, \sigma_{i,n,t,s}^+$	Dual variables of the BESS discharging-side power-headroom bounds.	€/MWh
$\gamma_{i,n,t,s}$	Dual variable of the SOC transition equation.	€/MWh
$\delta_{i,n,t,s}^-, \delta_{i,n,t,s}^+$	Dual variables of the lower and upper SOC energy bounds.	€/MWh
$\xi_{n,t,s}^-, \xi_{n,t,s}^+$	Dual variables of the lower and upper load-shedding bounds.	€/MWh
Parameters		
ω_s	Probability weight of activation scenario s , with $\sum_s \omega_s = 1$.	dimensionless
$\alpha_{t,s}^{up}$	Realized upward aFRR activation factor in time period t and scenario s , with $0 \leq \alpha_{t,s}^{up} \leq 1$.	dimensionless
$\alpha_{t,s}^{down}$	Realized downward aFRR activation factor in time period t and scenario s , with $0 \leq \alpha_{t,s}^{down} \leq 1$.	dimensionless
C_g	Marginal generation cost of generator g .	€/MWh
$VOLL$	Value of lost load for electricity demand.	€/MWh
$VOLL^{aFRR}$	Penalty value for unserved aFRR reserve demand.	€/MW
D_b^{aFRR}	aFRR reserve demand in block b . ^a	MW
$P_{g,t}^{max}$	Maximum available dispatch of generator g in time period t .	MW
$Limit_l$	Transmission capacity limit on line l .	MW
$PTDF_{l,n}$	Power transfer distribution factor from node n to line l .	dimensionless
$D_{n,t}^{el}$	Electricity demand at node n and time t . ^a	MW
η	One-way BESS charging and discharging efficiency, $0 \leq \eta \leq 1$.	dimensionless

^a Renamed relative to the original thesis notation for traceability: A_{up} is written as $P^{aFRR,up}$, A_{down} as $P^{aFRR,down}$, $Demand_b$ as D_b^{aFRR} , and $D_{n,t}$ as $D_{n,t}^{el}$.

^b The unit of $\lambda_{aFRR,b}$ is €/MW because it prices reserve capacity. This differs from $\lambda_{n,t,s}$, which

prices energy or power balance in €/MWh.

1 Introduction

Lower-Level Problem — Co-Optimized Spot + aFRR Market Clearing

Objective

$$\min \sum_s \omega_s \left[\sum_{t,g} C_g \cdot P_{g,t,s} + \sum_{n,t} VOLL \cdot P_{n,t,s}^{shed} \right] + \sum_b VOLL^{aFRR} \cdot P_b^{aFRR,def}$$

Subject to

$$\sum_{g \in G_n} P_{g,t,s} + \sum_i (P_{i,n,t,s}^{discharge} - P_{i,n,t,s}^{charge} - P_{i,n,t,s}^{aFRR,up} + P_{i,n,t,s}^{aFRR,down}) + P_{n,t,s}^{shed} - D_{n,t}^{el} = NetInjection_{n,t,s}, \quad \forall n, t, s \quad (\lambda_{n,t,s})$$

This defines nodal net injections from generation, storage dispatch, activated aFRR energy, and load shedding in each scenario.

$$\sum_n NetInjection_{n,t,s} = 0, \quad \forall t, s \quad (\lambda_{sys,t,s})$$

This enforces system-wide power balance by requiring total net injection to be zero in every time period and scenario.

$$\sum_{i,n} R_{i,n,b}^{aFRR} + P_b^{aFRR,def} \geq D_b^{aFRR}, \quad \forall b \quad (\lambda_{aFRR,b})$$

This clears enough aFRR reserve capacity to meet block-level reserve demand, with deficit slack preserving feasibility.

$$0 \leq R_{i,n,b}^{aFRR} \leq R_{i,n,b}^{offer}, \quad \forall i, n, b$$

This limits cleared aFRR capacity by the reserve volume offered by each storage unit.

$$0 \leq P_{g,t,s} \leq P_{g,t}^{max}, \quad \forall g, t, s \quad (\nu_{g,t,s}^-, \nu_{g,t,s}^+)$$

This keeps conventional and renewable generation within available capacity.

$$-Limit_l \leq \sum_n PTDF_{l,n} \cdot NetInjection_{n,t,s} \leq Limit_l, \quad \forall l, t, s \quad (\mu_{l,t,s}^-, \mu_{l,t,s}^+)$$

This represents network feasibility through PTDF-based transmission limits.

$$0 \leq P_{i,n,t,s}^{charge} + R_{i,n,b}^{aFRR} \leq X_{i,n}^{power}, \quad \forall i, n, t \in b, s \quad (\rho_{i,n,t,s}^-, \rho_{i,n,t,s}^+)$$

This reserves charging-side power headroom for aFRR while respecting the BESS power rating.

$$0 \leq P_{i,n,t,s}^{discharge} + \textcolor{blue}{R}_{i,n,b}^{aFRR} \leq X_{i,n}^{power}, \quad \forall i, n, t \in b, s$$

$$(\sigma_{i,n,t,s}^-, \sigma_{i,n,t,s}^+)$$

This reserves discharging-side power headroom for aFRR while respecting the BESS power rating.

$$\textcolor{blue}{P}_{i,n,t,s}^{aFRR,up} = \alpha_{t,s}^{up} \textcolor{blue}{R}_{i,n,b}^{aFRR}, \quad \forall i, n, t \in b, s$$

This fixes upward aFRR activation to the realized scenario activation factor times cleared reserve capacity.

$$\textcolor{blue}{P}_{i,n,t,s}^{aFRR,down} = \alpha_{t,s}^{down} \textcolor{blue}{R}_{i,n,b}^{aFRR}, \quad \forall i, n, t \in b, s$$

This fixes downward aFRR activation to the realized scenario activation factor times cleared reserve capacity.

$$SOC_{i,n,t,s} = SOC_{i,n,t-1,s} + \eta(P_{i,n,t,s}^{charge} + \textcolor{blue}{P}_{i,n,t,s}^{aFRR,down})$$

$$- \frac{1}{\eta}(P_{i,n,t,s}^{discharge} + \textcolor{blue}{P}_{i,n,t,s}^{aFRR,up}), \quad \forall i, n, t, s \quad (\gamma_{i,n,t,s})$$

This updates the BESS state of charge after market dispatch and upward or downward aFRR activation.

$$0 \leq SOC_{i,n,t,s} \leq X_{i,n}^{energy}, \quad \forall i, n, t, s \quad (\delta_{i,n,t,s}^-, \delta_{i,n,t,s}^+)$$

This keeps stored energy within the installed BESS energy capacity.

$$SOC_{i,n,0,s} = SOC_{i,n,24,s}, \quad \forall i, n, s$$

This imposes daily SOC consistency by requiring the terminal SOC to match the initial SOC.

$$0 \leq P_{n,t,s}^{shed} \leq D_{n,t}^{el}, \quad \forall n, t, s \quad (\xi_{n,t,s}^-, \xi_{n,t,s}^+)$$

This caps involuntary load shedding by the local electricity demand.

$$0 \leq \textcolor{blue}{P}_b^{aFRR,def} \leq \textcolor{blue}{D}_b^{aFRR}, \quad \forall b$$

This caps the aFRR deficit variable by total reserve demand in each block.

Exogenous parameters (fixed from upper level or given):

$$X_{i,n}^{power}, X_{i,n}^{energy}, \textcolor{blue}{R}_{i,n,b}^{offer}, C_g, P_{g,t}^{max}, D_{n,t}^{el}, \textcolor{blue}{D}_b^{aFRR},$$

$$Limit_l, PTDF_{l,n}, \eta, VOLL, \textcolor{blue}{VOLL}^{aFRR}, \alpha_{t,s}^{up}, \alpha_{t,s}^{down}, \omega_s$$